NATTY CARD NUTRITION

FREE GUIDE · V0.2 DRAFT

The Evidence-Graded Supplement Cheat Sheet

What works for natural lifters, and what's a **waste of money**.

A Strong **B** Moderate **C** Emerging **D** Not supported

For the outcome listed, healthy adults 18+. [n] = reference, p. 8 · † = NCN product planned (p. 8)

INGREDIENT	OUTCOME	GRADE	RESEARCH DOSE	TYPICAL LABEL PROBLEM	KEY SOURCE
WORTH IT: STRONG EVIDENCE					
Creatine monohydrate †	Strength & lean mass, with resistance training	A	**3–5 g/day.** Optional load: 20 g/day as 4 × 5 g for 5–7 days, then 3–5 g/day	Pricier "advanced" forms with no proven edge over monohydrate	ISSN 2017 [1]; Health Canada [2]
Protein (total daily)	Muscle gain, with resistance training	A	**1.4–2.0 g/kg/day** from food + powder; 20–40 g per meal, every 3–4 h	Less protein per scoop than the headline, or a huge scoop	ISSN 2017 [3]; Morton 2018 [4]
Caffeine	Endurance & many performance tests	A	Research: **3–6 mg/kg**, ~60 min before. Canadian NHP limit: **100–200 mg/dose, max 400 mg/day**	Total caffeine per serving not stated; stacked stimulants	ISSN 2021 [5]; Health Canada [6]
USEFUL IN THE RIGHT CONTEXT					
Beta-alanine	Hard efforts of ~1–4 min	B	**4–6 g/day** for 2–4+ weeks, split doses (NHP max 6.4 g/day, 3.2 g/dose)	One scoop is far below the daily research dose	ISSN 2015 [7]; Health Canada [6]
Dietary nitrate (beetroot)	Endurance (time to exhaustion)	B	**310–560 mg nitrate** (5–9 mmol), 2–3 h before	Lists grams of beet powder, not mg of nitrate	IOC 2018 [8]; Poon 2025 [9]
Sodium bicarbonate	Hard efforts of ~45 s to 8 min	B	**0.2–0.3 g/kg**, 60–180 min before. GI upset is common	Large sodium load; stomach issues	ISSN 2021 [10]
Sodium / electrolytes	Replacing sweat losses in long (over ~1–2 h), sweaty sessions	B	Drink to limit large body-weight losses. Long sessions: **~20–30 mEq/L sodium** (~460–690 mg per litre)	"Hydration" mixes with little sodium	ACSM 2007 [11]; Del Coso 2016 [36]; Cosgrove 2013 [37]
MAYBE, NOT YET: NO DOSE SUGGESTED					
Citrulline malate	Reps to failure (one session)	C	*No dose suggested*	Dose or citrulline:malate ratio not disclosed	Vårvik 2021 [12]; Aguiar 2022 [13]
HMB †	Strength in people new to resistance training. **Trained lifters: D**	C	*No dose suggested (licensed products use 3–6 g/day CaHMB; not a recommendation)*	Sold as "anti-catabolic"	Rowlands 2009 [30]; Jakubowski 2020 [14]; Sanchez-Martinez 2018 [31]
Ashwagandha	Strength & recovery	C	*No dose suggested. See safety note, page 7*	Extract type and standardisation not stated	Bonilla 2021 [15]; Health Canada [16]
Omega-3 (fish oil)	Blood markers of muscle damage after hard exercise	C	*No dose suggested*	EPA + DHA per capsule not stated	Xin 2021 [17]
SKIP IT: NOT SUPPORTED					
BCAAs	Muscle gain, protein already adequate	D	—	Sold as essential when total protein covers it	Plotkin 2021 [19]; Wolfe 2017 [20]
Glutamine	Muscle gain or strength	D	—	Sold as a muscle builder or "anti-catabolic"	Candow 2001 [21]; Ramezani Ahmadi 2019 [32]
Tribulus terrestris	Testosterone or strength	D	—	"Natural testosterone" marketing	Qureshi 2014 [22]
ZMA	Testosterone, if not deficient	D	—	"Anabolic" or sleep-gains claims	Koehler 2009 [23]
"Test boosters" †	Testosterone	D	—	Proprietary blends; claims with no data	Clemesha 2020 [24]
Magnesium	Sleep in healthy adults	D	— Too little evidence in healthy adults; the only meta-analysis studied older adults with sleep problems	Form and elemental amount unclear	Mah 2021 [18]

Grades apply to the outcome listed, in healthy adults, for the form and dose studied. Not medical advice. A/B doses are what research used, kept within Health Canada natural health product (NHP) monograph limits where one exists. Talk to a health professional before starting any supplement.

BEFORE YOU BUY

The 5-Minute Label Check

Five questions. Any supplement. If it fails two or more, put it back on the shelf.

1 **Is every dose disclosed?**
A "proprietary blend" lists the ingredients but not how much of each. Treat it as a red flag: you can't compare it with the research dose.

2 **Does each key ingredient hit the research dose?**
Compare the label with page 1. For example: creatine 3–5 g/day; beta-alanine 4–6 g/day; nitrate in mg, not grams of beet powder.

3 **Is there an 8-digit Health Canada NPN?**
In Canada, a Natural Product Number (NPN) means Health Canada licensed the product for sale. Look the number up in the Licensed Natural Health Products Database [28].

4 **Is the third-party testing real?**
Look for a programme such as NSF Certified for Sport or Informed Sport. Then find the exact product in the certifier's own online database. A logo on the tub isn't proof.

5 **What's the cost per *effective* dose?**
Not per scoop. Count how many research doses the container really holds.

Worked example (made-up numbers)

A tub holds **300 g** of creatine monohydrate. Research dose: **5 g**.

300 ÷ 5 = 60

effective doses. Cost per effective dose = **price ÷ 60**. If one dose takes 2 scoops, count it that way.

Quick red flags

- proprietary blend
- "clinically proven" (no citation)
- "anabolic"
- "natural steroid"
- "test booster"
- before/after photos
- no NPN or lot number

Where to check (free, official databases)

Health Canada NPN
health-products.canada.ca/lnhpd-bdpsnh/
Search the 8 digits only [28]

NSF Certified for Sport
nsfsport.com/certified-products/

Informed Sport
sport.wetestyoutrust.com

OUR METHOD

How We Grade (A–D)

We grade each ingredient for a **specific outcome** in **healthy adults**. "Caffeine for endurance" and "caffeine for a one-rep max" are different questions, and they can get different grades. ISSN's 2021 caffeine position stand found the most consistent benefits for aerobic endurance [5].

GRADE	MEANING	WHAT WE USUALLY NEED TO SEE
A	**Strong**	At least one good systematic review or meta-analysis of human randomized controlled trials (RCTs), plus consistent RCTs in the relevant population. Expert position stands (e.g. ISSN, ACSM/Dietitians of Canada) support a grade but aren't required.
B	**Moderate**	Several good human RCTs showing benefit, but smaller effects, narrower contexts or some inconsistency.
C	**Emerging**	At least one human controlled trial on the outcome showing a benefit, but small, mixed, unreplicated or industry-only.
D	**Not supported**	Good human trials show no meaningful benefit, or there's too little human evidence to judge. On this sheet, only magnesium for sleep is D for lack of evidence; our Methodology page labels that case **I** (Insufficient).

We downgrade for

Industry-funded studies without independent replication · doses far above what products contain · surrogate outcomes only (e.g. a hormone level rather than strength or muscle) · very short trials. Testimonials and marketing never raise a grade.

Counts as evidence

- Human randomized controlled trials
- Systematic reviews and meta-analyses
- Position stands from expert bodies (ISSN, IOC, ACSM), as support

Doesn't count

- Testimonials and influencer results
- Animal-only or cell-only data
- Mechanism-only claims ("it raises nitric oxide, so it must build muscle")
- "Clinically proven" with no citation

Why the outcome matters: one ingredient, two answers

B **Sodium bicarbonate for hard efforts of ~45 s to 8 min.** Pooled analyses show a benefit [10][29].

D **Sodium bicarbonate for muscle strength.** An umbrella review found no significant difference from placebo [29].

A grade isn't a product claim. It's an evidence summary for one outcome. It says nothing about safety, drug interactions or anti-doping status, and a product using a different form or a lower dose doesn't inherit the grade. Full method: [site URL]/methodology/

BUDGET-AWARE

The Start-Here Stack

Three A-grade basics, in order. Most natural lifters never need more than this.

STEP 0

Foundations first

No supplement makes up for missed training, short sleep or too little food. Get these right first: a progressive resistance-training plan, consistent sleep, enough total calories for your goal, and enough protein.

STEP 1

Hit your protein target, with food first

- Target **1.4–2.0 g/kg/day** in total [3]. In a meta-analysis of 49 RCTs, gains in fat-free mass levelled off at around **1.6 g/kg/day**; the authors note up to about 2.2 g/kg/day may help some people [4].
- Spread it out: about **20–40 g** (or ~0.25 g/kg) per meal, every 3–4 hours [3].
- Use protein powder **only if food doesn't get you there**. It's food in a tub, not magic.

STEP 2

Creatine monohydrate, 3–5 g/day

- Take it every day; consistency matters more than timing. Start at 3–5 g/day (stores fill over about 3–4 weeks), or load with 20 g/day as 4 × 5 g for 5–7 days, then 3–5 g/day [1][2].
- Buy plain creatine **monohydrate**. Other forms haven't been shown to work better [1].

STEP 3

Caffeine (optional), dosed with care

- Research doses: 3–6 mg/kg about 60 minutes before training [5].
- Canadian NHP limits for adult products: **100–200 mg per dose, max 400 mg/day**. Keep your total from all sources (coffee and energy drinks count) to 400 mg/day or less [6].
- Start low. Skip it late in the day if it hurts your sleep.

Do the math for your body weight (example: 80 kg lifter)

128 g
protein/day at 1.6 g/kg (80 × 1.6). About 4 meals of ~32 g.

3–5 g
creatine monohydrate/day. Same for most adults.

200 mg
caffeine per dose, max. 3 mg/kg would be 240 mg, but licensed Canadian products give no more than 200 mg per dose, so start there.

Skip until the basics are covered

BCAAs · glutamine · "test boosters" · fat burners · any multi-ingredient product with a proprietary blend.

SAVE YOUR MONEY

The Skip List

D-grade for the outcome listed. Common buys the evidence doesn't support.

- **D** **BCAAs**, if you already eat enough protein. When daily protein is adequate, adding BCAAs or leucine hasn't been shown to improve strength or muscle growth [19][20].
- **D** **Glutamine for muscle gain.** A 6-week RCT using very high doses, and a meta-analysis of 25 trials in athletes, found no effect on strength or lean mass [21][32].
- **D** **Tribulus terrestris.** A systematic review found it ineffective for raising testosterone in humans [22].
- **D** **ZMA for testosterone** in people who aren't deficient. A high-dose zinc/magnesium product did not change serum testosterone in men who trained regularly [23].
- **D** **HMB for trained lifters.** Meta-analyses found no meaningful strength or body-composition benefit in trained people [14][31].
- **D** **"Test boosters".** A review of 50 top-selling products and 109 ingredients [24]:

90%
of products claimed to boost testosterone

24.8%
of ingredients had data showing a rise

18.3%
had data showing no change

10.1%
had data showing a **decrease** (61.5% had no data at all; categories overlap)

Marketing words to watch for

"anabolic" · "natural steroid" · "test booster" · "hormone optimiser" · "clinically proven" · "doctor-formulated" · "proprietary blend"

"Natural steroid alternatives" don't exist

The phrase sells a feeling, not a result. We aren't aware of any legal supplement shown in human trials to build muscle on anything like the scale of anabolic steroids. Even creatine, the best-supported option, adds roughly 1 kg of lean mass over a few months of training, on average [35]. In "booster" products, most ingredients have no testosterone data at all [24]. Products that do produce drug-like results are often found to contain undeclared drugs, which is illegal and risky. Natural progress comes from training, food, sleep and time.

A **Spend it here instead**

The start-here stack on page 4: hit your protein target, take creatine monohydrate 3–5 g/day, and add caffeine only if you want it.

TOP OF THE A LIST

Creatine in 60 Seconds

What it does

Your muscles store creatine as phosphocreatine, which quickly recycles ATP (your cells' immediate energy) during short, hard efforts. More stored creatine lets you do a bit more work per set, and that adds up over months of training [1].

Dose

3–5 g/day

Creatine monohydrate, every day. Optional load: 20 g/day as 4 × 5 g for 5–7 days, then 3–5 g/day [1][2].

Myths, answered

MYTH **"It causes hair loss."** The idea comes from one 2009 study of college rugby players (16 completed it) that measured a hormone (DHT), not hair. Its DHT result has not been replicated [25][26]. A 2025 12-week RCT measured hair directly and found no differences in DHT or hair outcomes versus placebo [27].

MYTH **"It damages your kidneys."** In healthy people, studies lasting up to several years haven't found kidney harm at recommended doses [1][25][33]. Creatine can slightly raise creatinine on a blood test [34], which can make a routine kidney estimate look lower than it really is. Tell your clinician you take it before bloodwork [1][25]. If you have a kidney condition, ask a health professional first [2].

MYTH **"It's just water weight."** Early weight gain during loading includes extra water stored *in the muscle* (about 0.5–1.0 L in early studies) [1]. Health Canada's label wording says you may gain weight [2]. Longer-term gains in lean mass come with training [1].

What Health Canada allows a label to say

Creatine monohydrate products can claim to increase lean muscle mass **with resistance training**, and improve strength/power in **repeated, brief, high-intensity activity**, for adults 18+ [2].

Plant-based?

Vegetarians tend to have lower muscle creatine stores, so they may see a larger rise in muscle creatine when they supplement [1].



Want the full evidence?

Creatine, Decoded is our first ebook, coming soon. You're already on our email list, so you'll hear first.

Join the ebook waitlist: `[link]`

READ THIS FIRST

Safety Basics

Evidence-based also means knowing when a supplement isn't for you.

♥ **Talk to a health professional first**
If you're pregnant or breastfeeding, under 18, have a medical condition (including kidney disease), or take medication.

♠ **More isn't better**
Stay within label directions and Health Canada limits. For caffeine, that means no more than 400 mg/day for healthy adults, from all sources [6].

♣ **Ashwagandha: know the warning signs**
Health Canada's 2024 safety review found no definitive link to liver injury, but couldn't rule one out. Stop use and seek medical advice if you get yellow skin or eyes, dark urine, abdominal pain, unusual tiredness, nausea or loss of appetite [16].

♦ **Tested athletes: you're responsible**
For everything you take. Use products certified by a third-party programme (e.g. NSF Certified for Sport, Informed Sport) and check the exact product in the certifier's database.

✓ **"Natural" doesn't mean safe or tested**
In Canada, check for an 8-digit NPN [28].

! **Had a reaction?**
Stop using the product, get medical help if needed, and report it to Health Canada: canada.ca, search "report a side effect natural health product" [38]. You can also tell the company that sells the product; licensed companies must report serious reactions to Health Canada.

Bring this to your pharmacist or doctor

- The product name, its NPN and the dose you plan to take.
- Every other supplement, medication and caffeine source you use.
- Any condition you're managing, and any upcoming bloodwork.

Health disclaimer

This guide is general information and education, not medical advice. It doesn't diagnose, treat, cure or prevent any disease. Talk to a qualified health professional before you start a supplement.

Who's behind this

Natty Card Nutrition is a sports nutrition brand based in Kitchener-Waterloo, Ontario, founded by Jason Cimbron.
`[Founder bio, credentials, evidence reviewer]`

How we make money

This PDF has **no affiliate links**. Our site and emails include some affiliate links: if you buy through one, we may earn a commission at no extra cost to you. Commissions never change our grades. NCN is also developing its own capsule supplements (not for sale yet). Where we have or plan a product, we say so and use the same rubric. On this sheet, those rows are marked † (creatine, HMB, "test boosters").

Methodology: [site URL]/methodology/ · Reviews: [site URL]/reviews/ · Disclosures: [site URL]/affiliate-disclosure/

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